

Ph134, Lecture 10

- **Coherence time and coherence length**
- **Fourier transform relationship between $s(\omega)$ and $\text{Re} [g^{(1)}(\tau)]$**
- **Illustrations**
- **Example: Doppler-broadened atomic transitions**

Source with continuous wavelength distribution?

$$E^{(+)}(t) = \int_0^{\infty} \mathcal{E}^{(+)}(\omega) e^{-i\omega t} d\omega$$

angular frequency
bandwidth (width of
the distribution in
angular frequency
space) $\gg \frac{2\pi}{T_{det}}$

$$I_{IN} = \int_0^{\infty} \tilde{I}(\omega) d\omega$$

⇒ So it makes sense to talk about the input power as having a certain fraction $s(\omega)d\omega$ in each small frequency interval. Then:

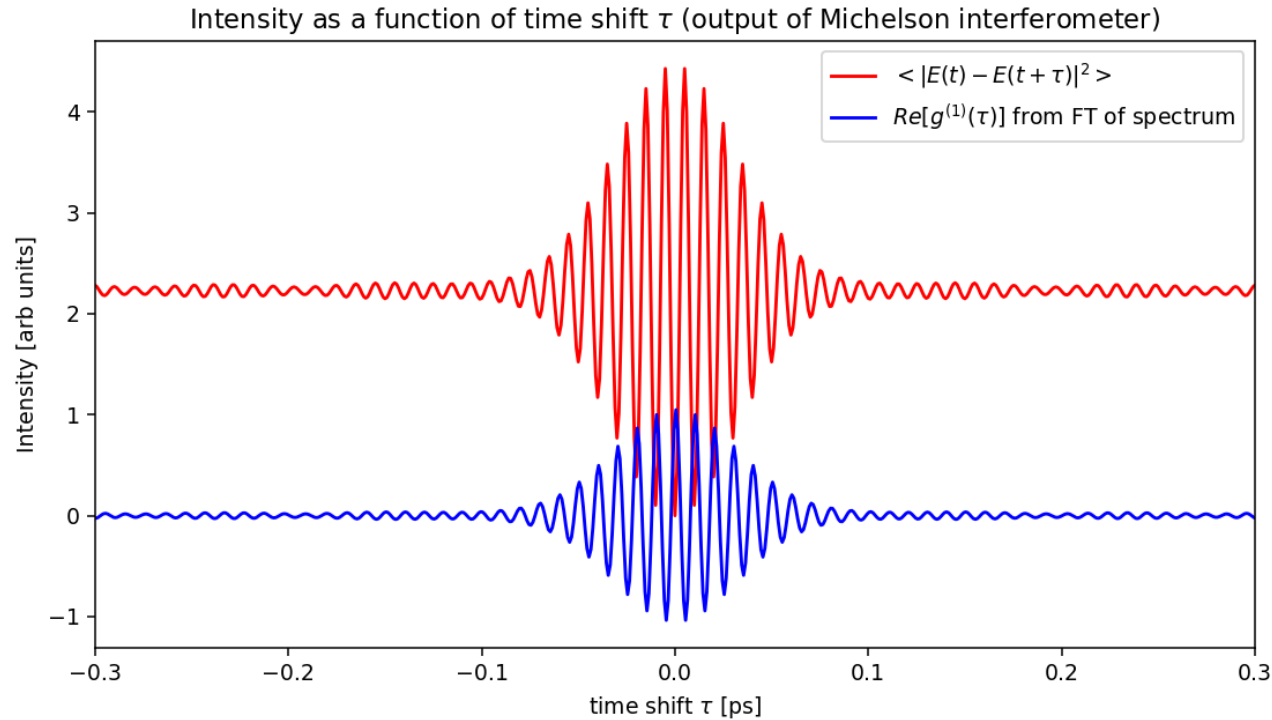
$$g^{(1)}(\tau) = \int_0^{\infty} s(\omega) e^{-i\omega\tau} d\omega$$

$$\text{Re } g^{(1)}(\tau) = \int_0^{\infty} s(\omega) \cos \omega\tau d\omega$$

Fringe visibility depends on the individual monochromatic interference patterns being in sync with each other or not.

Once patterns get out of sync, no “recovery” at larger τ . Limited range around $\tau=0$ where coherence is present / fringes observed.

Coherence time and coherence length:



$$\tau_{coh} \equiv \int_{-\infty}^{\infty} (\text{Re}[g^{(1)}(\tau)])^2 d\tau \quad \ell_{coh} = c\tau_{coh}$$

$$(\tau_{coh})(\text{source bandwidth}) \sim 1$$

Source with continuous wavelength distribution?

$$E^{(+)}(t) = \int_0^{\infty} \mathcal{E}^{(+)}(\omega) e^{-i\omega t} d\omega$$

width Ω of the
distribution in
angular
frequency space

$$\gg \frac{2\pi}{T_{\text{detection}}}$$

$$I_{IN} = \int_0^{\infty} \tilde{I}(\omega) d\omega$$

⇒ So it makes sense to talk about the input power as having a certain fraction $s(\omega)d\omega$ in each small frequency interval. Then:

$$g^{(1)}(\tau) = \int_0^{\infty} s(\omega) e^{-i\omega\tau} d\omega$$

$$\text{Re } g^{(1)}(\tau) = \int_0^{\infty} s(\omega) \cos \omega\tau d\omega$$

And by Fourier transform
pair relationship,

$$s(\omega) = \frac{2}{\pi} \int_0^{\infty} \text{Re } [g^{(1)}(\tau)] \cos \omega\tau d\tau$$

Real electric field and two-sided power spectrum:

$$E^{(+)}(t) = \int_0^{\infty} \mathcal{E}^{(+)}(\omega) e^{-i\omega t} d\omega$$

$$\Rightarrow E(t) = \int_0^{\infty} \mathcal{E}^{(+)}(\omega) e^{-i\omega t} d\omega + \int_0^{\infty} [\mathcal{E}^{(+)}(\omega)]^* e^{i\omega t} d\omega$$

$$= \int_{\omega=0}^{\infty} \mathcal{E}^{(+)}(\omega) e^{-i\omega t} d\omega + \int_{-\omega=-\infty}^0 [\mathcal{E}^{(+)}(-(-\omega))]^* e^{-i(-\omega)t} d(-\omega)$$

$$= \int_{-\infty}^{\infty} \mathcal{E}(\omega) e^{-i\omega t} d\omega \quad \mathcal{E}(\omega) \equiv \begin{cases} \mathcal{E}^{(+)}(\omega) & \omega > 0 \\ 2\mathcal{E}^{(+)}(\omega) & \omega = 0 \\ [\mathcal{E}^{(+)}(-\omega)]^* & \omega < 0 \end{cases}$$

Correspondingly, can talk about 2-sided power spectrum

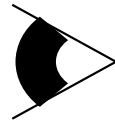
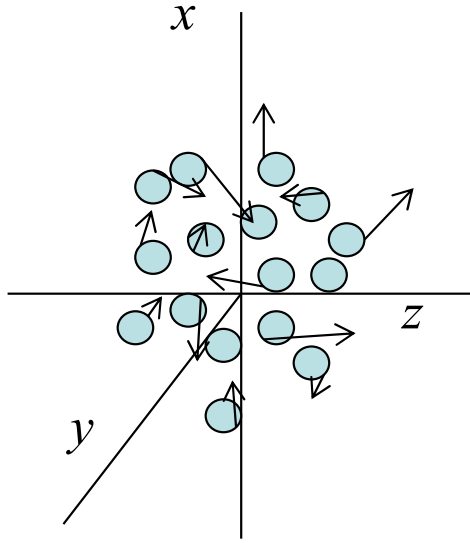
$$s_{\leftrightarrow}(\omega) \equiv \begin{cases} s(\omega)/2 & \omega > 0 \\ s(\omega) & \omega = 0 \\ s(-\omega)/2 & \omega < 0 \end{cases}$$

Examples of $E(t)$, $s \leftrightarrow (\omega)$, and $\text{Re } g^{(1)}(\tau)$:

Coherence and Source Bandwidth (Fourier!) Jupyter Notebook

Source with continuous wavelength distribution?

Doppler-broadened atomic transition



Atoms emit light at angular frequency ω_0 in their own rest frames.

From atom with v_z , observer receives light at angular frequency

$$\omega \approx \omega_0 \left(1 + \frac{v_z}{c} \right)$$

Thermal spread in v_z values $\rightarrow$ observed spread in ω .

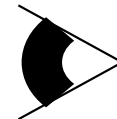
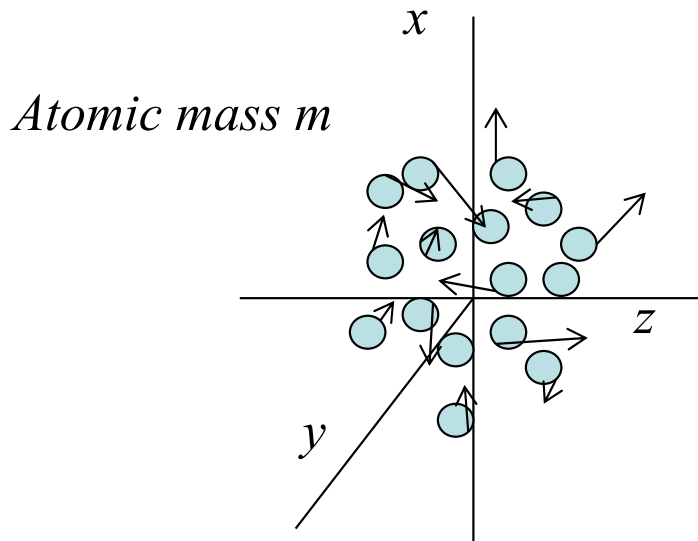
$s(\omega)$ from Doppler-broadened atomic transition

$$P(v_z) = \frac{1}{\sqrt{2\pi \left(\frac{k_B T}{m}\right)}} e^{-v_z^2 / 2 \left(\frac{k_B T}{m}\right)}$$

$$\omega \approx \omega_0 \left(1 + \frac{v_z}{c}\right)$$
$$s(\omega) d\omega = P(v_z) dv_z$$

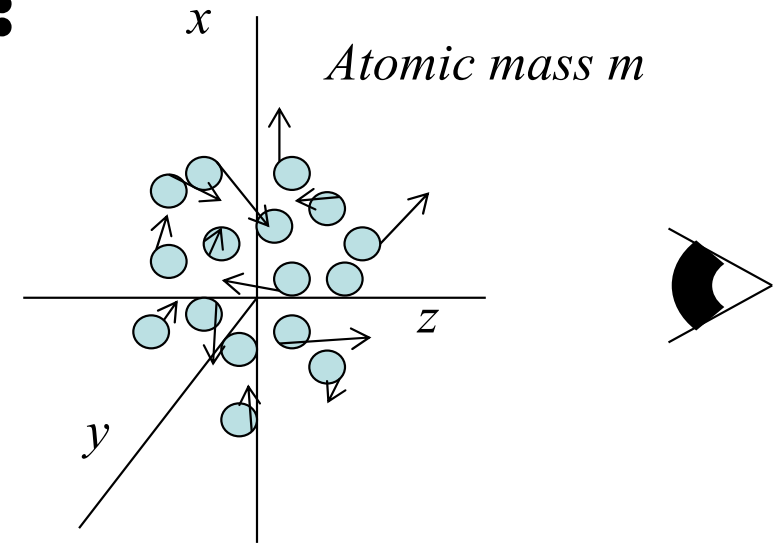
$$s(\omega) = \frac{1}{\sqrt{2\pi\sigma_\omega^2}} e^{-(\omega - \omega_0)^2 / 2\sigma_\omega^2}$$

$$\sigma_\omega^2 = \left(\frac{\omega_0}{c}\right)^2 \frac{k_B T}{m}$$



Doppler broadening examples:

$$s(\omega) = \frac{1}{\sqrt{2\pi\sigma_\omega^2}} e^{-(\omega - \omega_0)^2 / 2\sigma_\omega^2}$$
$$\sigma_\omega^2 = \left(\frac{\omega_0}{c}\right)^2 \frac{k_B T}{m}$$



Examples:

*He-Ne laser (633nm), Ne mass 20 amu, $T \sim 370$ K:

$$FWHM = 2\pi * 1.5 \text{ GHz, or } 0.02 \text{ \AA}$$

*low-pressure Hg lamp (546nm), Hg mass 200 amu, $T \sim 370$ K:

$$FWHM = 2\pi * 540 \text{ MHz, or } 0.0054 \text{ \AA}$$

*solar H line (656nm), H mass 1 amu, $T \sim 5000$ K:

$$FWHM = 2\pi * 24 \text{ GHz, or } 0.34 \text{ \AA due to Doppler alone}$$

Interferometer output for a Doppler-broadened source?

From Fourier transform knowledge or by other means, find that $V(\tau)$ is also Gaussian (centered at $\tau = 0$).